ENERGY CALCULATION FOR RESIDENTIAL AIR-CONDITIONER FOR INDIAN SUB-CONTINENT

PARAMETERS

Enviromental

- HUMIDEX
- COOLING DEGREE DAYS

PENETRATION

- NUMBER OF AC
- NUMBER OF HOUSEHOLD
  - AVERAGE HOUSEHOLD SIZE
  - TOTAL FERTILITY RATE
- HOUSEHOLD INCOME

EFFICIENCY FACTOR

- ISSER RATING
  - NUMBER OF UNIT SOLD PER STAR
- OPERATING HOURS

FINAL ENERGY CALCULATION

$$EC_{VARIABLE} = \sum_{i=1}^{5} \frac{1.3\ tone(kW) \times Unit\ Sold\ for\ i^{th}\ Star \times Operating\ Hours}{ISEER\ Rating\ for\ i^{th}\ Star} + EC_{FIXED} = \sum_{i=1}^{5} \frac{1.3\ tone(kW) \times Unit\ Sold\ for\ i^{th}\ Star \times Operating\ Hours}{ISEER\ Rating\ for\ i^{th}\ Star}$$

1) ISSER

(Standards & Labelling)S&L

ISEER values follows the current growth trend i.e., ratcheting up of MEPS at a rate of 2.76% annually till 2029-30

Rate of efficiency improvements in ISEER with a value of 1.7% annually between 2030-31 and 2039-40

Rate of efficiency improvements in ISEER with a value of 1% annually beyond 2040-41

ISEER value in 2046-47 would be 6.37

Average deterioration in efficiency of the units occurs at a rate of 2% annually

India Cooling Action Plan (ICAP)

Ratcheting up of MEPS happens at double the rate of current S&L programme, i.e., ISEER improves at a rate of 5.52% annually till 2029-30

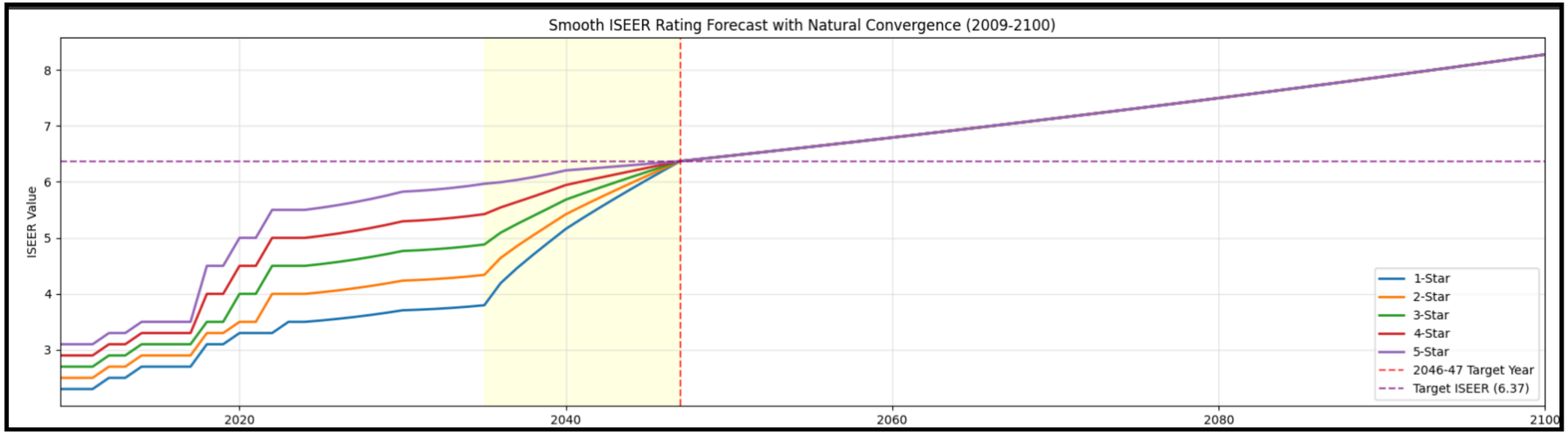
Rate of efficiency improvements in ISEER with a value of 3.4% annually between 2030-31 and 2039-40

Rate of efficiency improvements in ISEER with a value of 2% annually beyond 2040-41

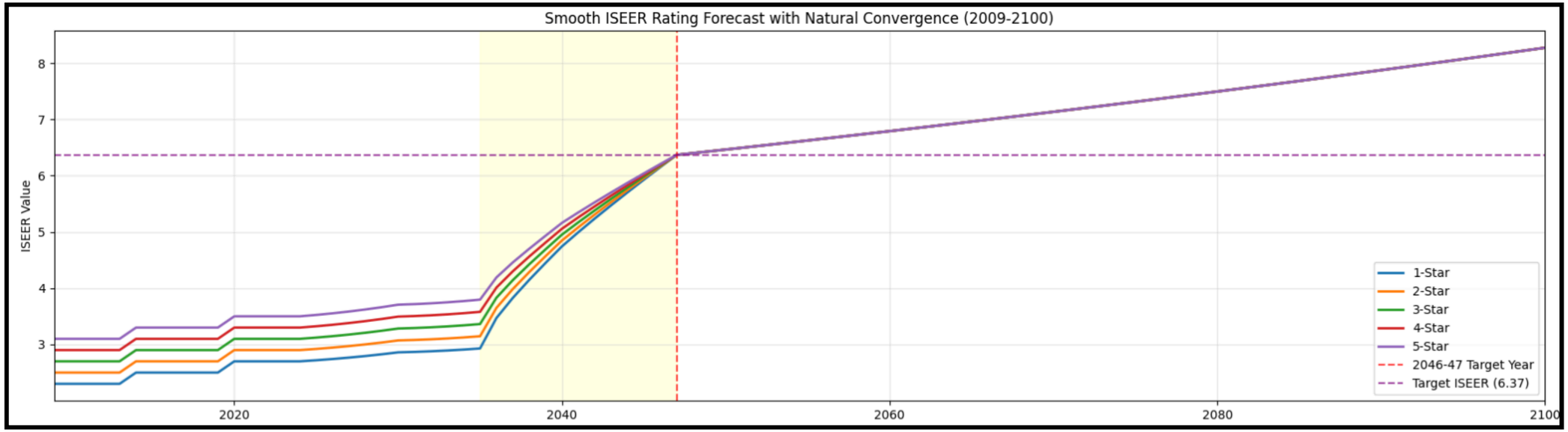
ISEER value in 2046-47 would be 10.78

Average deterioration in efficiency of the units occurs at a rate of 2% till 2024-25 and gradually improves to 0.5% considering the efforts under ICAP implementation around training and certification of technicians and consumer awareness.

(Standards & Labelling)



Fixed Speed ACs

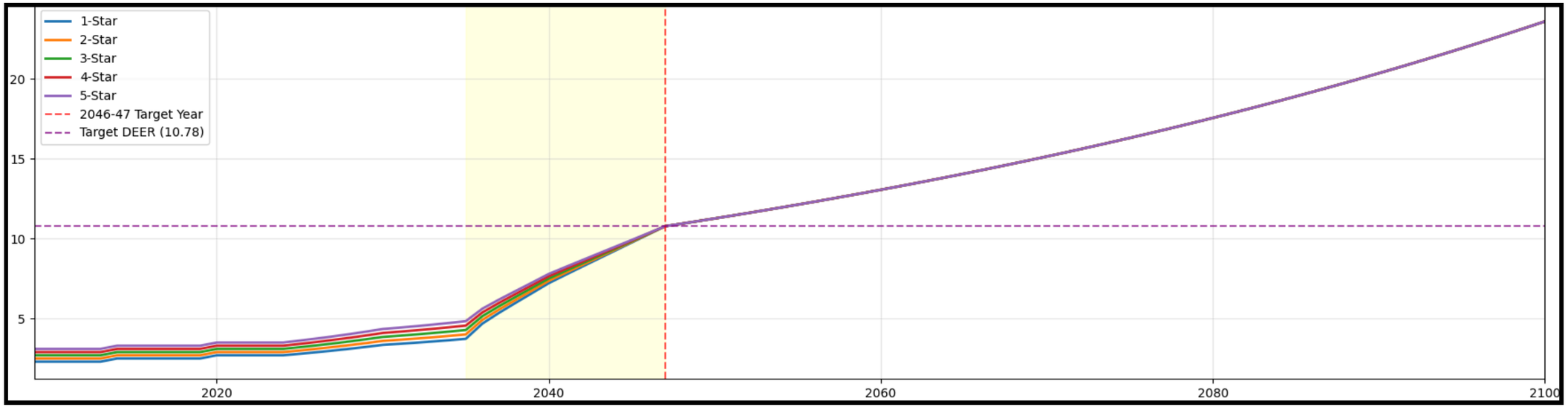


Variable Speed ACs

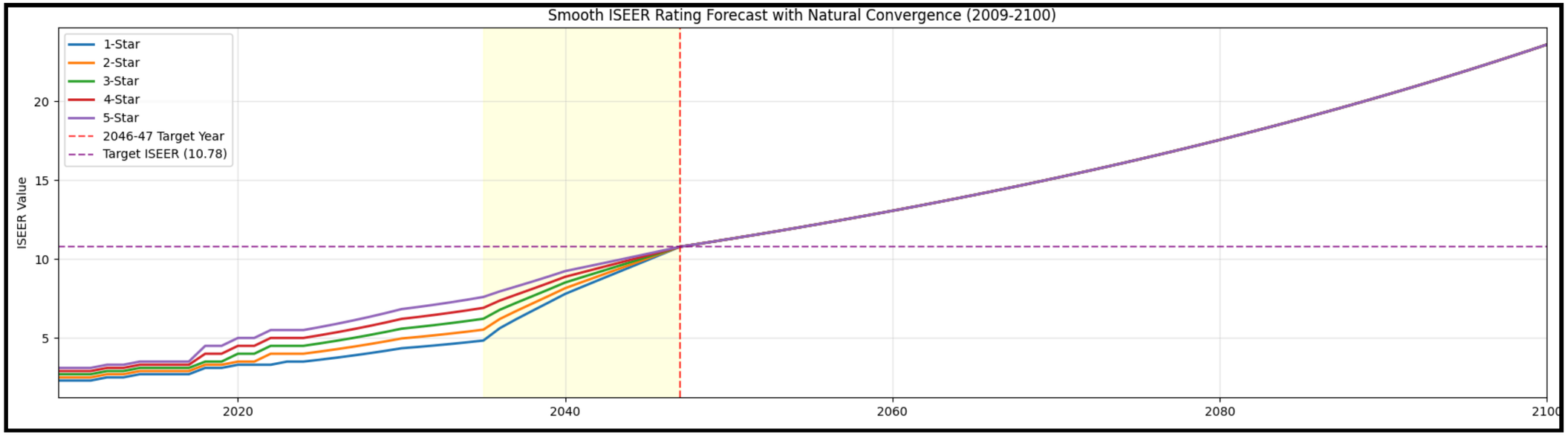
Verification for 2046:

1-Star	6.37
2-Star	6.37
3-Star	6.37
4-Star	6.37
5-Star	6.37

India Cooling Action Plan (ICAP)



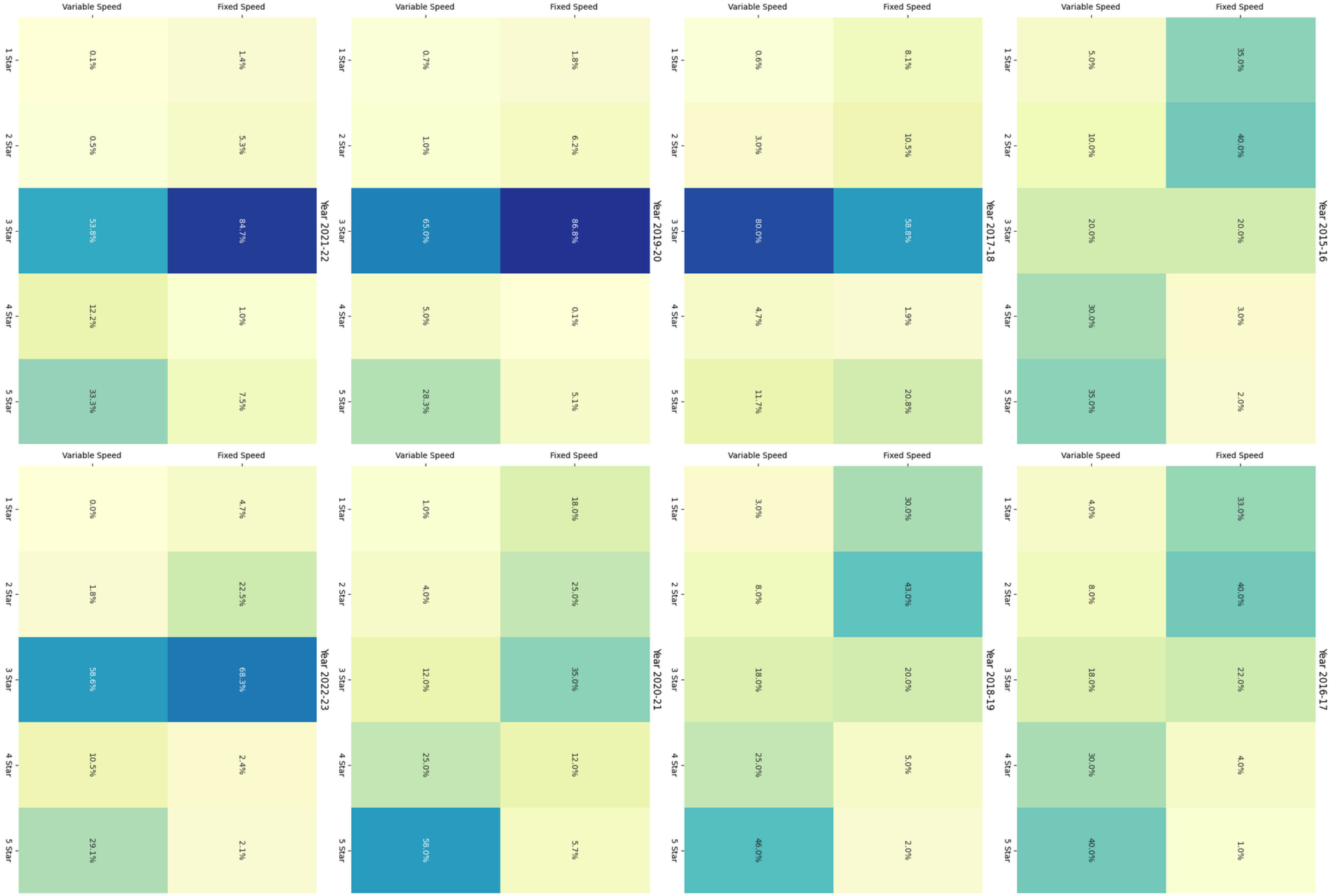
Fixed Speed ACs



Variable Speed ACs

Verification for 2046:
1-Star 10.78
2-Star 10.78
3-Star 10.78
4-Star 10.78
5-Star 10.78

NUMBER OF UNIT SOLD PER STAR



Fixed Speed ACs (%)

Year	1 Star	2 Star	3 Star	4 Star	5 Star
2015-16	35	40	20	3	2
2016-17	33	40	22	4	1
2017-18	30	42	20	5	3
2018-19	30	43	20	5	2
2019-20	23	35	27	10	5
2020-21	18	25	35	12	10
2021-22	18	20	40	15	7
2022-23	15	15	50	15	5
A	5.3	6.7	56.5	21.4	10.1

Variable Speed ACs (%)

Year	1 Star	2 Star	3 Star	4 Star	5 Star
2015-16	5	10	20	30	35
2016-17	4	8	18	30	40
2017-18	3	8	18	28	43
2018-19	3	8	18	25	46
2019-20	2	5	15	25	53
2020-21	1	4	12	25	58
2021-22	1	3	10	24	62
2022-23	1	2	10	23	64
A’	1	2	7.38	19.4	70.3

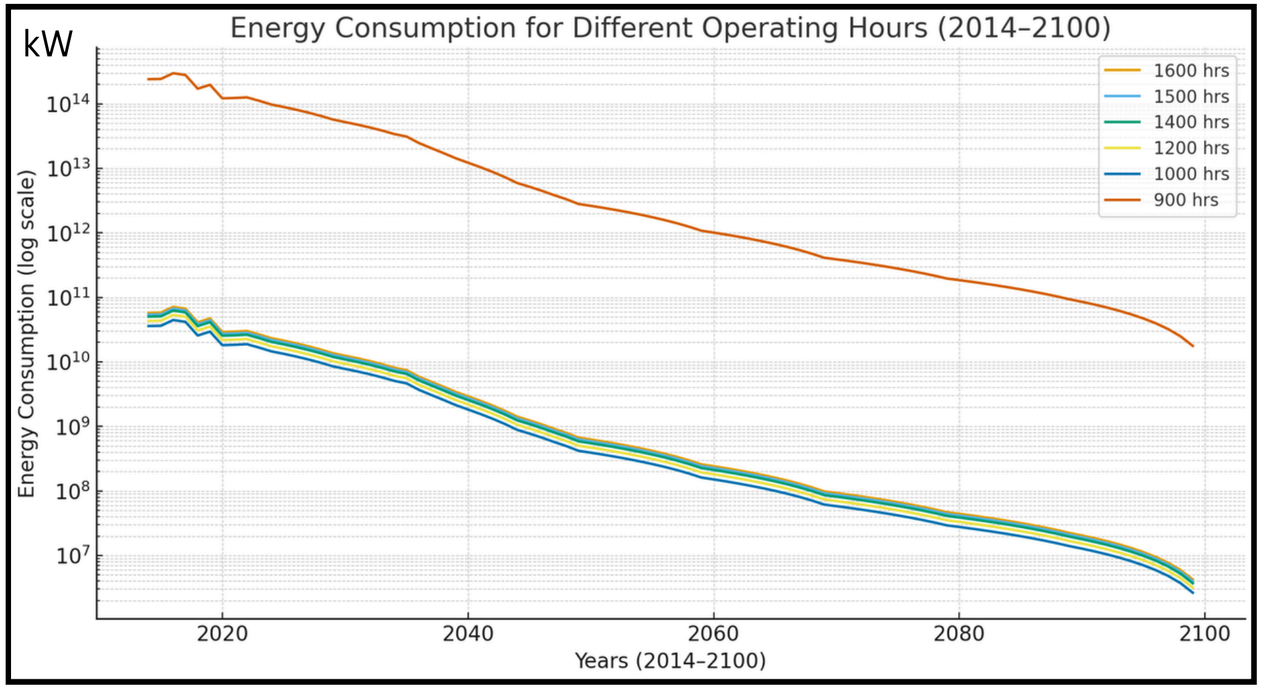
$$\textbf{Star based Unit Distribution(Fixed speed)} = \sum_{i=1}^{5} A_i \cdot i^{th}\ star\ units$$

$$\textbf{Star based Unit Distribution(Variable speed)} = \sum_{i=1}^{5} A'_i \cdot i^{th}\ star\ units$$

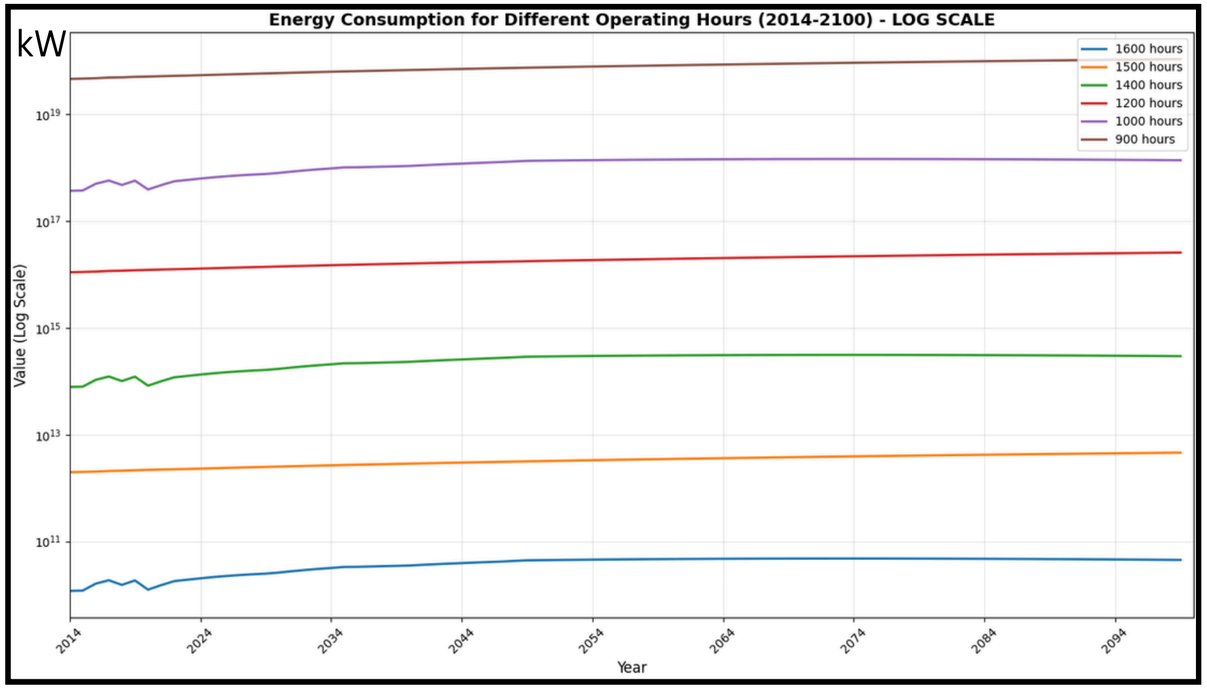
A = percentage of total unit sale coefficient for fixed speed AC units

A' = percentage of total unit sale coefficient for Variable speed AC units

ENERGY CONSUMPTION FOR DIFFERENT OPERATING HOURS



S&L

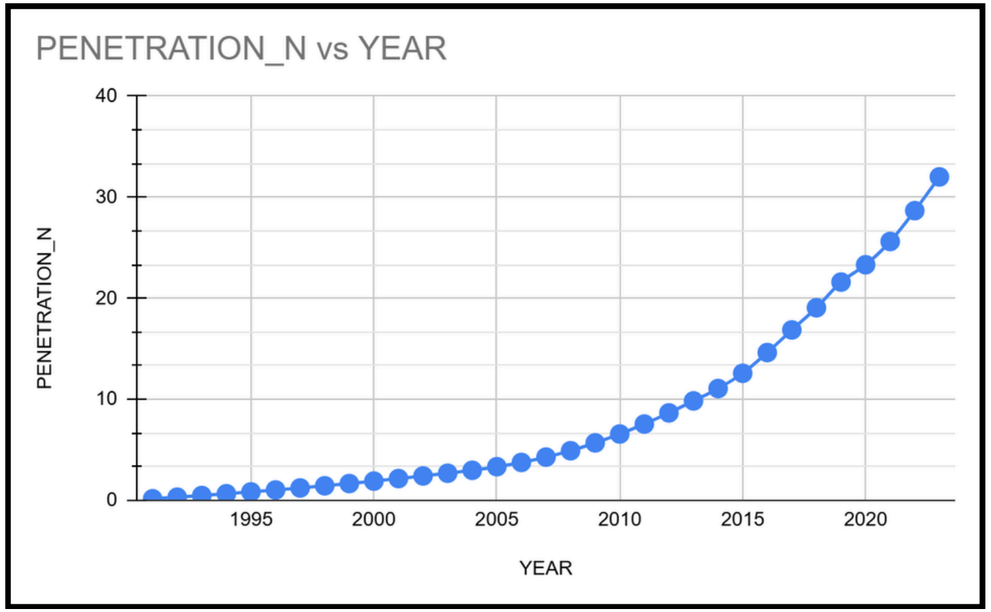


ICAP

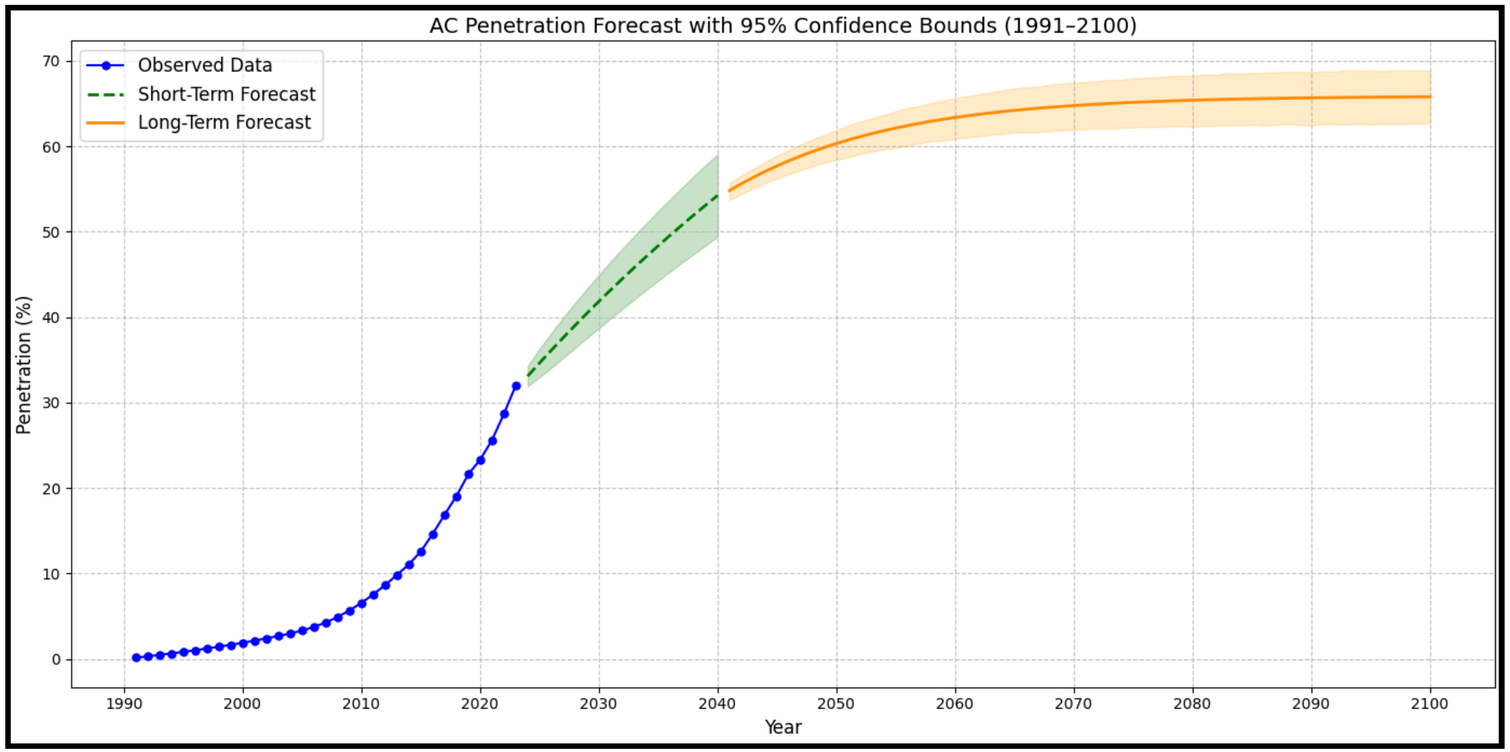
GROWTH ANALYSIS BY OPERATING HOURS

Operating Hours	Start (2014)	End (2100)	Absolute Increase	Growth %	Annual Growth Rate %
1600 hours	1.17e+10	4.47e+10	3.30e+10	281.33	3.27
1500 hours	1.97e+12	4.59e+12	2.62e+12	132.67	1.54
1400 hours	7.81e+13	2.95e+14	2.17e+14	277.77	3.22
1200 hours	1.10e+16	2.58e+16	1.47e+16	133.54	1.55
1000 hours	3.71e+17	1.39e+18	1.02e+18	274.20	3.19
900 hours	4.65e+19	1.09e+20	6.25e+19	134.56	1.56

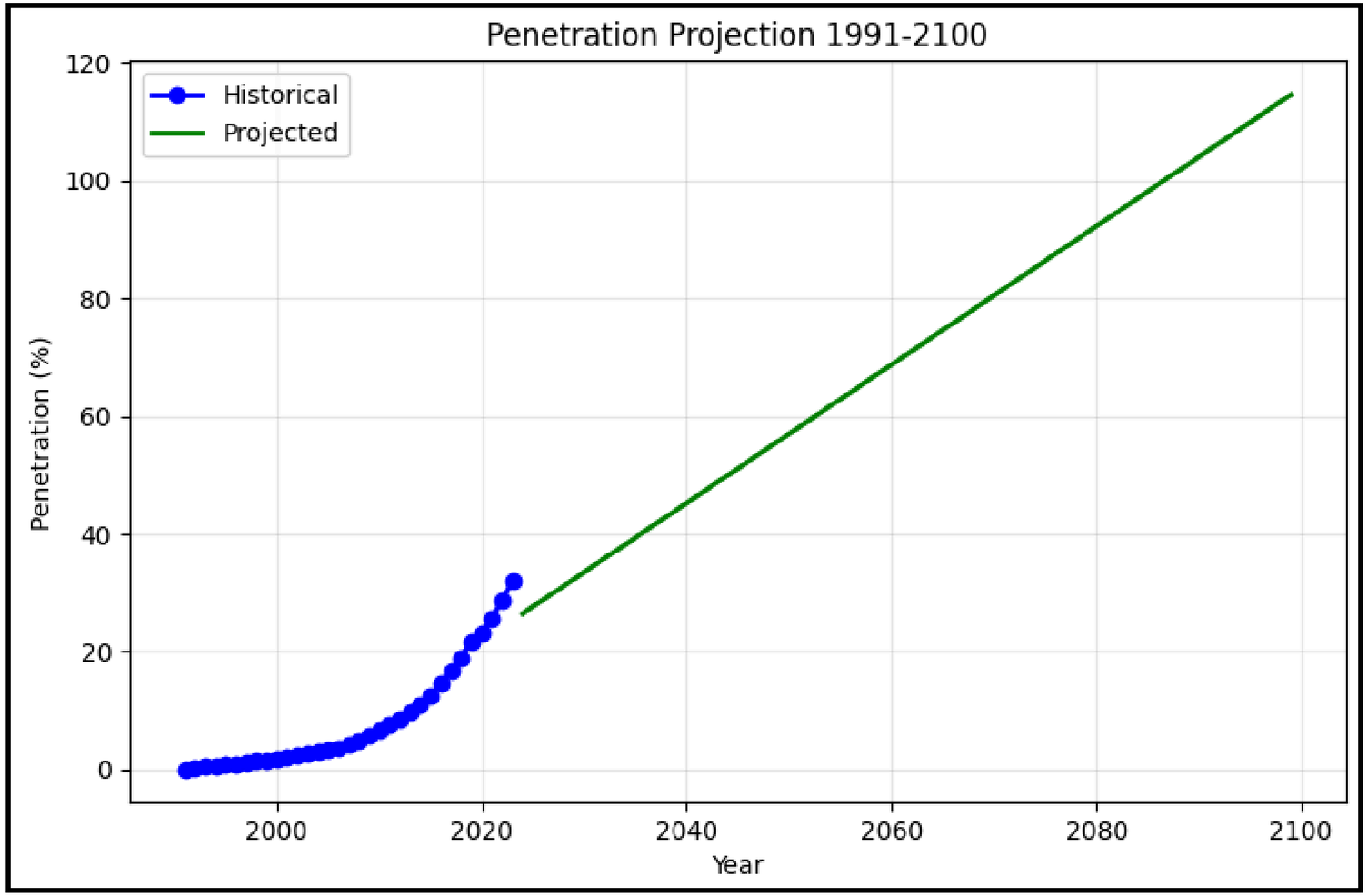
PENETRATION



$$\text{Penetration} = \frac{\textit{Number of Ac Units}}{\textit{Total Households}}$$



Year	Forecast (%)	Lower Bound (%)	Upper Bound (%)
1991	0.15	0.15	0.15
1992	0.31	0.31	0.31
1993	0.48	0.48	0.48
1994	0.65	0.65	0.65
1995	0.84	0.84	0.84
1996	1.03	1.03	1.03
1997	1.23	1.23	1.23
1998	1.44	1.44	1.44
1999	1.66	1.66	1.66
2000	1.90	1.90	1.90
2001	2.16	2.16	2.16
2002	2.42	2.42	2.42
2003	2.68	2.68	2.68
2004	2.97	2.97	2.97
2005	3.33	3.33	3.33
2006	3.75	3.75	3.75
2007	4.28	4.28	4.28
2008	4.91	4.91	4.91
2009	5.69	5.69	5.69
2010	6.57	6.57	6.57
2011	7.55	7.55	7.55
2012	8.66	8.66	8.66
2013	9.84	9.84	9.84
2014	11.07	11.07	11.07
2015	12.58	12.58	12.58
2016	14.63	14.63	14.63
2017	16.87	16.87	16.87
2018	19.06	19.06	19.06
2019	21.60	21.60	21.60
2020	23.32	23.32	23.32
2021	25.62	25.62	25.62
2022	28.67	28.67	28.67
2023	32.01	32.01	32.01
2024	33.11	31.93	34.28
2025	34.64	32.98	36.30
2026	36.14	34.11	38.17
2027	37.61	35.26	39.95
2028	39.05	36.43	41.67
2029	40.46	37.59	43.33
2030	41.85	38.74	44.95



$$Penetration = 4.8908 + 0.0044 \cdot Income - 0.0037 \cdot CDD_{RCP8.5}$$

$$Penetration = 4.8908 + 0.0044 \cdot Income - 0.0037 \cdot CDD_{RCP4.5}$$

$$Penetration = 2.1382 + 0.0045 \cdot Income - 0.0019 \cdot CDD_{humidex}$$

Year	Predicted Penetration_HUMIDEX	Predicted Penetration_RCP_8.5	Predicted Penetration_RCP_4.5
2024	26.45	26.32	26.25
2030	33.49	32.87	33.04
2040	45.22	43.07	44.12
2050	56.95	52.36	54.74
2060	68.69	62.28	64.93
2070	80.43	72.32	75.49
2080	92.17	82.36	82.22
2090	103.92	92.41	92.18
2099	114.49	101.44	107.25

VALIDATION

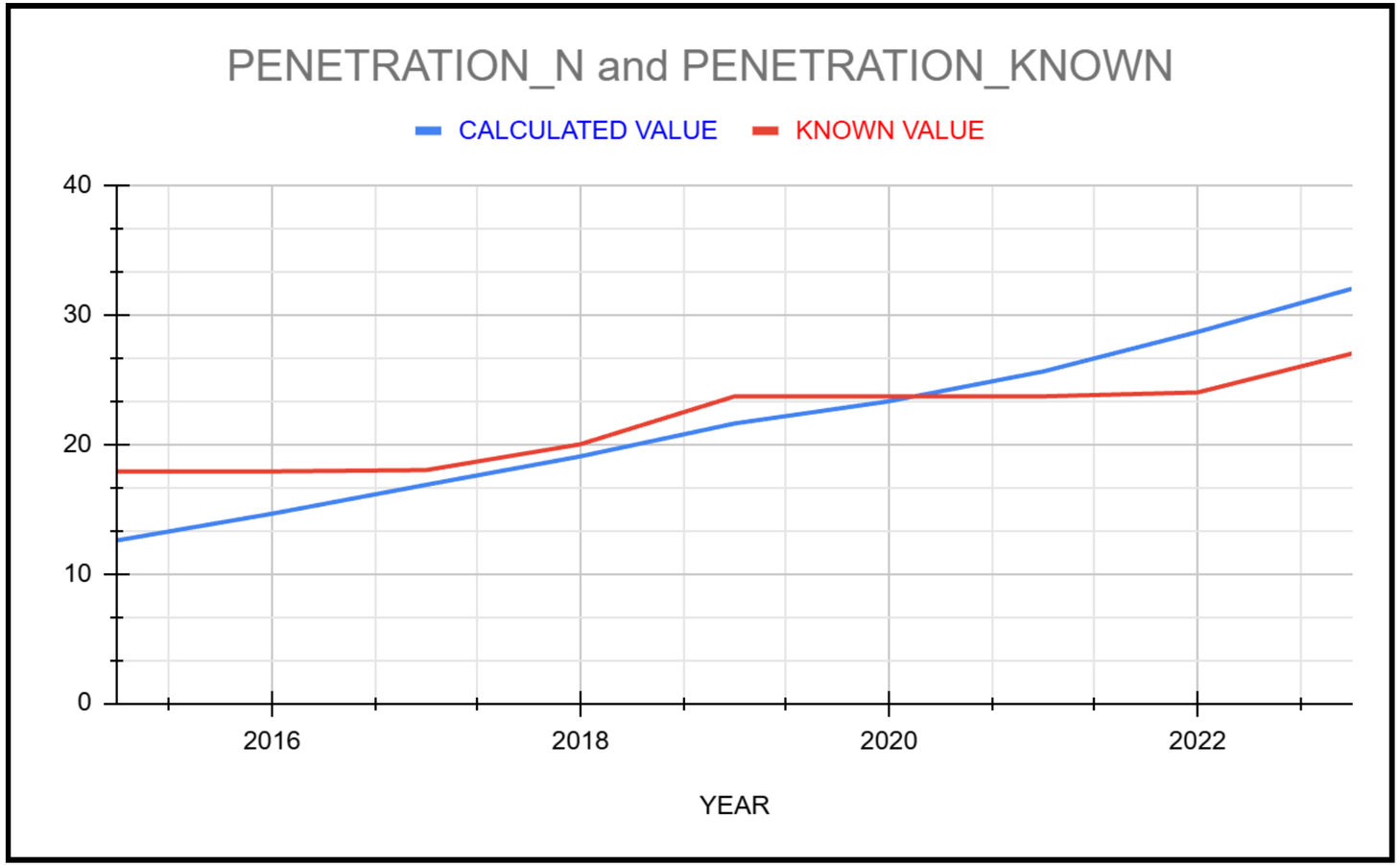


Fig.2.2. Penetration known Vs Calculated Value

2015	12.57578381	17.9
2016	14.62973811	17.9
2017	16.8659402	18
2018	19.06225478	20
2019	21.60320027	23.7
2020	23.31652458	23.7
2021	25.62107204	23.7
2022	28.67051388	24
2023	32.0119449	27
2050	60.32	45

Mean Absolute Error (MAE) = 2.75
Uncertainty = ≈ 12.6%

NFHS DATA

Table 2.11 Household possessions

Percentage of urban, rural, and total households and *de jure* population possessing various household goods and means of transport, and percentage with a bank/post office account, health insurance/financing scheme, or a BPL card, India, 2019-21

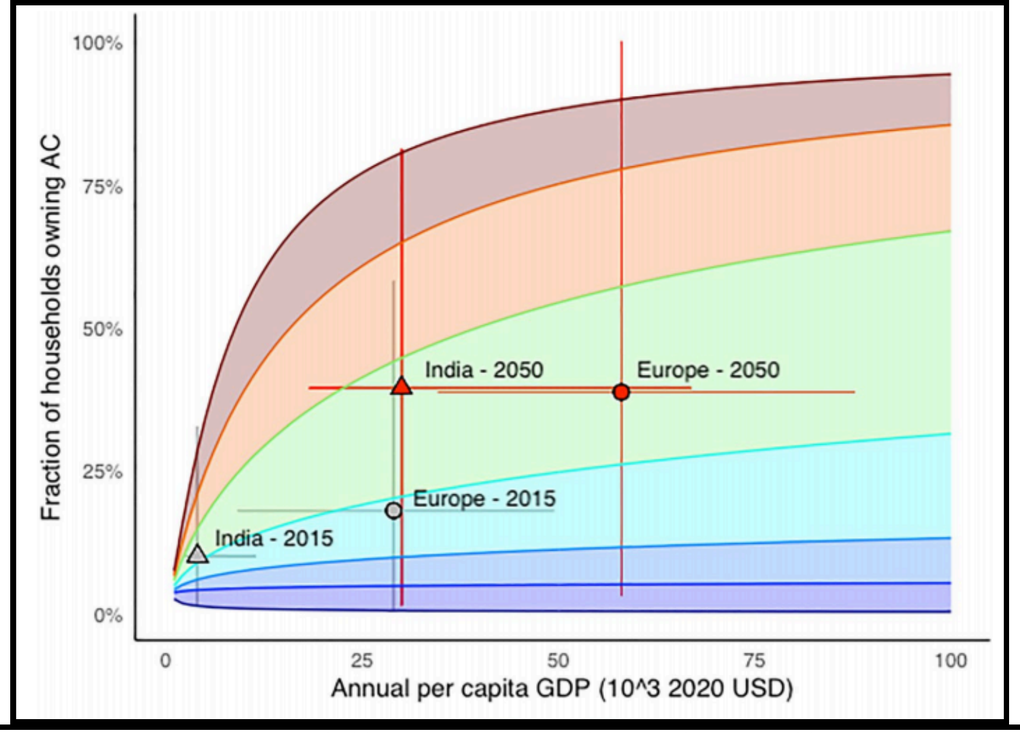
Household possessions	Urban	Rural	Total	*De jure* population
Household goods				
Mattress	85.8	66.1	72.6	74.2
Pressure cooker	88.8	57.9	68.2	69.8
Chair	90.5	80.1	83.5	84.5
Cot or bed	89.5	89.4	89.4	90.5
Table	74.2	51.3	58.9	60.3
Electric fan	96.4	84.3	88.3	89.1
Radio or transistor	6.7	4.1	4.9	4.9
Television (black and white)	2.2	2.3	2.3	2.4
Television (colour)	86.0	57.1	66.7	67.7
Any television	86.8	58.4	67.8	68.9
Sewing machine	34.0	22.7	26.4	30.2
Mobile telephone	96.7	91.5	93.3	95.5
Landline telephone	4.6	1.1	2.3	2.2
Internet	64.6	41.0	48.8	53.0
Computer	19.3	4.4	9.3	9.5
Refrigerator	63.4	25.2	37.9	38.9
Air conditioner/cooler	39.5	15.8	23.7	25.4
Washing machine	36.1	9.0	18.0	18.6
Watch or clock	90.3	70.7	77.2	78.3
Water pump	22.3	17.3	19.0	20.2
Thresher	0.6	1.7	1.4	1.7
Tractor	0.7	3.9	2.9	3.8
None of the above	0.1	0.6	0.4	0.3

Table 2.8 Household possessions

Percentage of urban, rural, and total households and *de jure* population possessing various household goods and means of transport, and percentage with a bank/post office account, health scheme/health insurance, or a BPL card, India, 2015-16

Household possessions	Urban	Rural	Total	*De jure* population
Household goods				
Mattress	82.3	58.4	66.7	67.7
Pressure cooker	83.6	42.2	56.6	57.4
Chair	86.5	70.7	76.2	76.9
Cot or bed	88.5	88.3	88.4	89.6
Table	72.1	46.5	55.4	56.3
Electric fan	95.1	69.1	78.2	78.2
Radio or transistor	10.3	7.0	8.1	8.4
Television (black and white)	3.1	3.5	3.3	3.7
Television (colour)	86.0	51.5	63.5	64.1
Any television	87.0	53.5	65.2	65.9
Sewing machine	33.5	19.0	24.0	26.9
Mobile telephone	96.1	87.3	90.4	92.9
Landline telephone	7.8	1.9	3.9	3.7
Internet	20.4	5.7	10.9	11.2
Computer	19.2	3.6	9.0	9.0
Refrigerator	54.2	16.4	29.6	30.1
Air conditioner/cooler	32.9	9.9	17.9	18.9
Washing machine	28.8	5.5	13.6	13.7
Watch or clock	90.8	71.4	78.1	79.3
Water pump	21.5	14.9	17.2	18.3
Thresher	0.6	1.9	1.5	1.8
Tractor	0.7	3.4	2.5	3.3
None of the above	0.2	1.2	0.8	0.6

Fig.2.3 NFHS



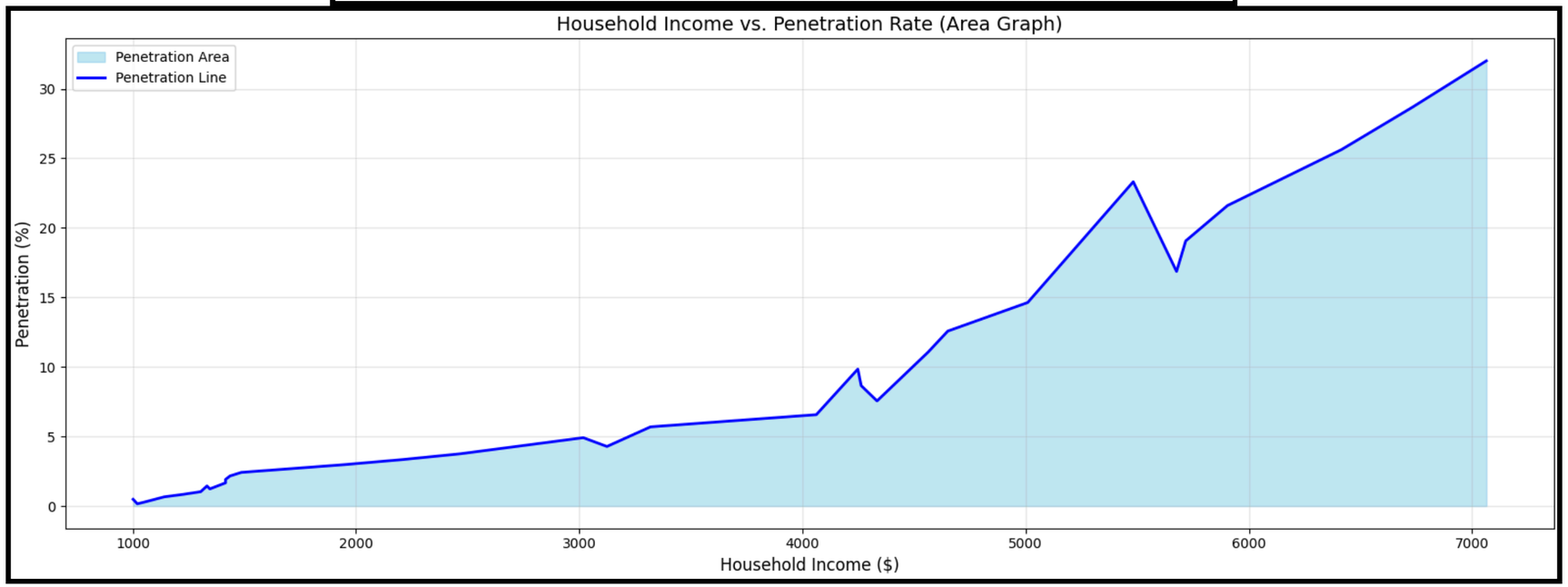


Fig.2.4 Income Vs Penetration

3.CDD

3.1 CDD Calculation

- **Using ASHRAE formula for calculating CDD[]**

According to the ASHRAE daily mean temperature method, the daily degree-days is the difference between the daily mean temperature, Td and base temperature, Tb. The calculations for and daily cooling degree-days, CDDd are given:

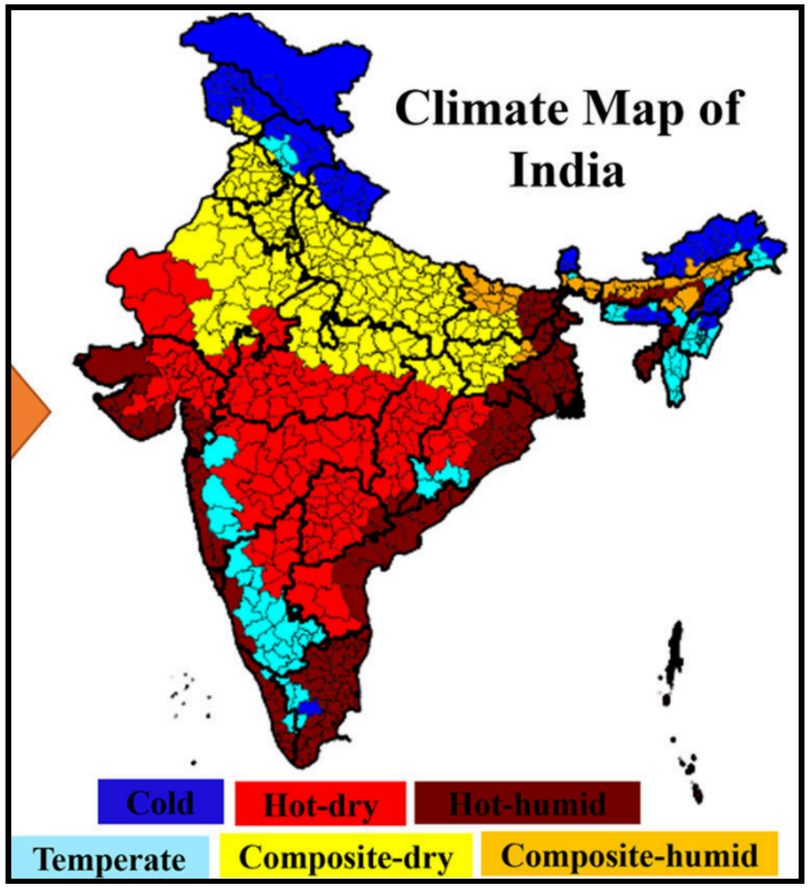
$$CDD_d = (T_d - T_b)^+$$

$$T_d = \frac{T_{max} + T_{min}}{2}$$

Td is calculated from daily maximum and minimum temperatures, Tmax and Tmin respectively:

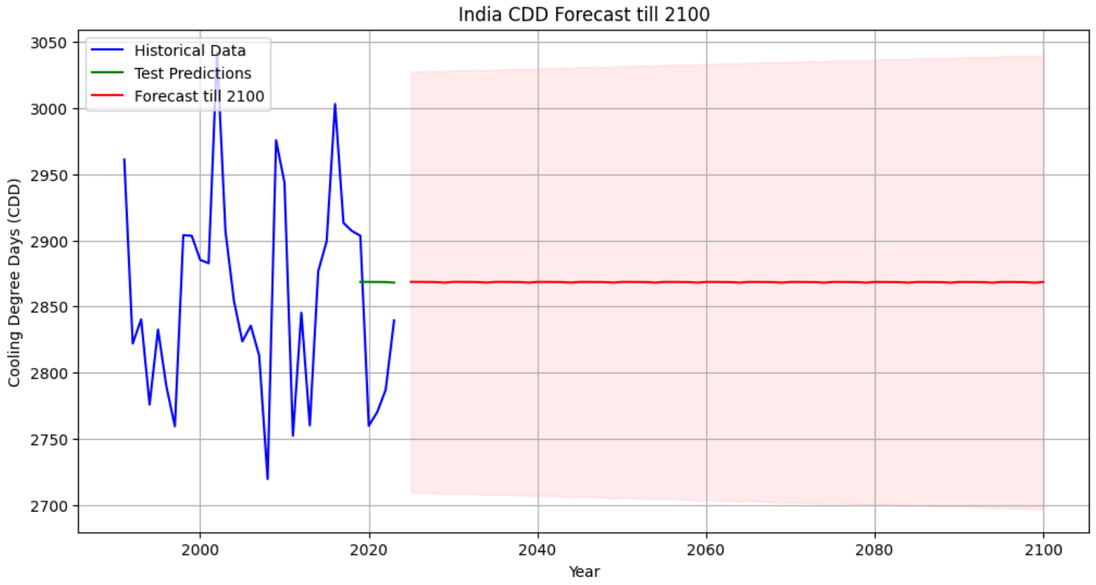
DATASET : https://power.larc.nasa.gov/data-access-viewer/ (For T_max and T_min daily data)

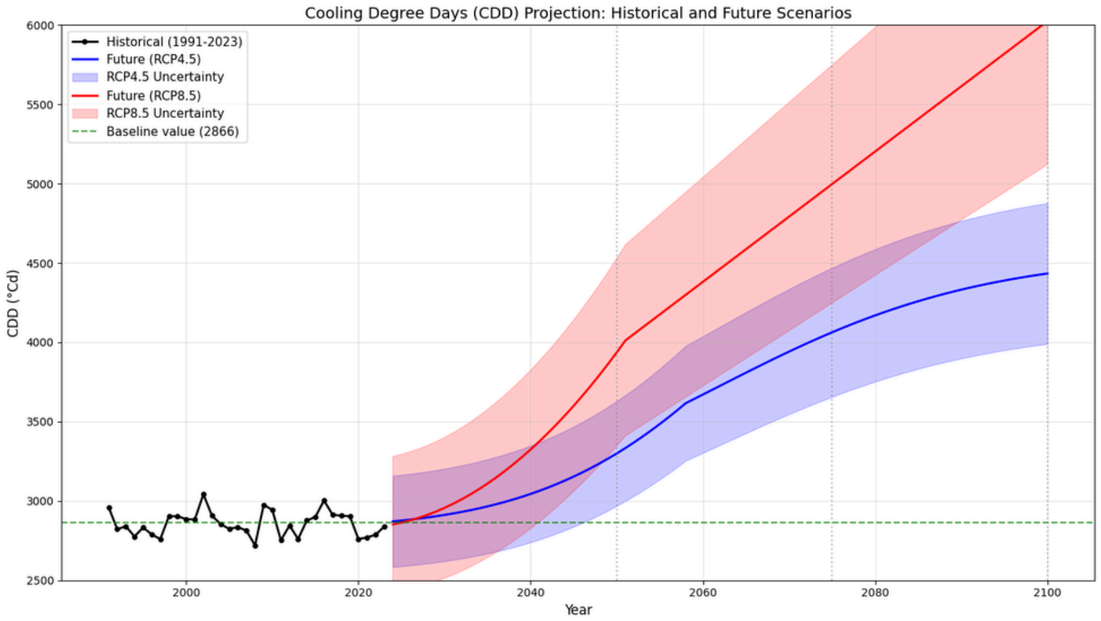
CLIMATIC ZONES



Climate Zone	Zone Code	Estimated coverage	Primary region
Hot- Dry	1B	30%	Rajasthan ,Western Gujrat
Hot-Humid	1A	30%	Costal parts of India , parts of southern India
Composite-humid	3B	4%	Eastern plains
Composite-dry	3A	14%	Northern Plains , Central Plains
Temperate	3C	9%	Southern Elevated area
Cold	5A	13%	Himalyan regions, Kashmir

$$CDD_{INDIA} = 0.3 \cdot CDD_{HOT-DRY} + 0.3 \cdot CDD_{HOT-HUMID} + 0.14 \cdot CDD_{COMPOSITE-DRY} + 0.04 \cdot CDD_{COMPOSITE-HUMID} + 0.09 \cdot CDD_{TEMPERATE} + 0.13 \cdot CDD_{COLD}$$





Decade	RCP4.5_Range	RCP4.5_%Inc	RCP8.5_Range	RCP8.5_%Inc
2020s	2789.2 (Historical)	-2.7	-	-
2030s	2667-3260	3.4	2637-3568	8.2
2040s	2833-3462	9.8	3043-4118	24.9
2050s	3123-3817	21.1	3530-4775	44.9
2060s	3415-4174	32.4	3882-5252	59.3
2070s	3644-4454	41.3	4231-5724	73.7
2080s	3824-4673	48.2	4580-6196	88.0
2090s	3944-4820	52.9	4929-6669	102.3
2100	3990-4877	54.7	5121-6928	110.2

HUMIDEX Based CDD CALCULATION

$$HUMIDEX = T_C + \frac{5}{9} \cdot \left(\frac{e_{RH}}{100} - 10 \right)$$

$$e_{RH} = \frac{RH}{100} \cdot e_s$$

$$e_s = 6.112 \cdot e^{\left(\frac{17.62 \cdot T_C}{243.12 + T_C} \right)}$$

$$CDD = max(0, Humidex - T_{base})$$

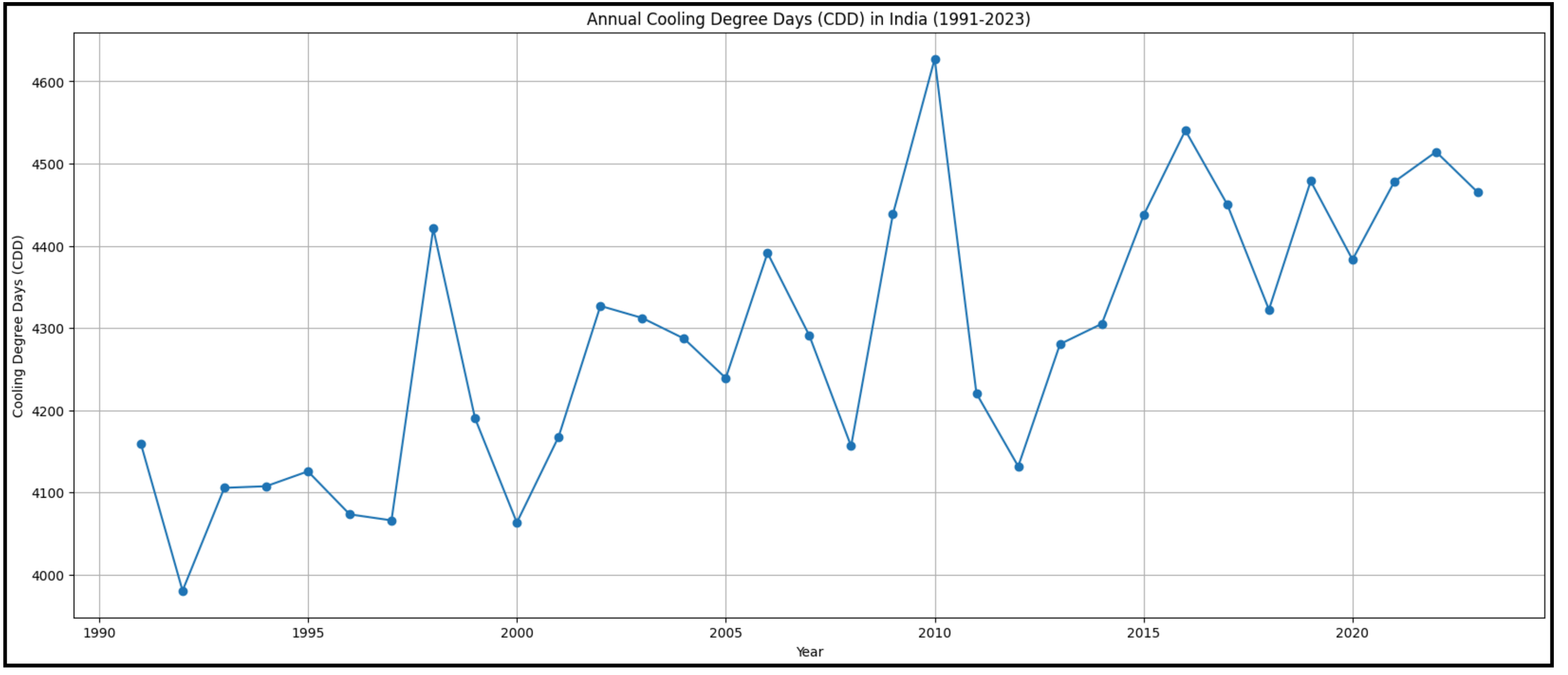
$T_C =$ *temperature in Celsius*

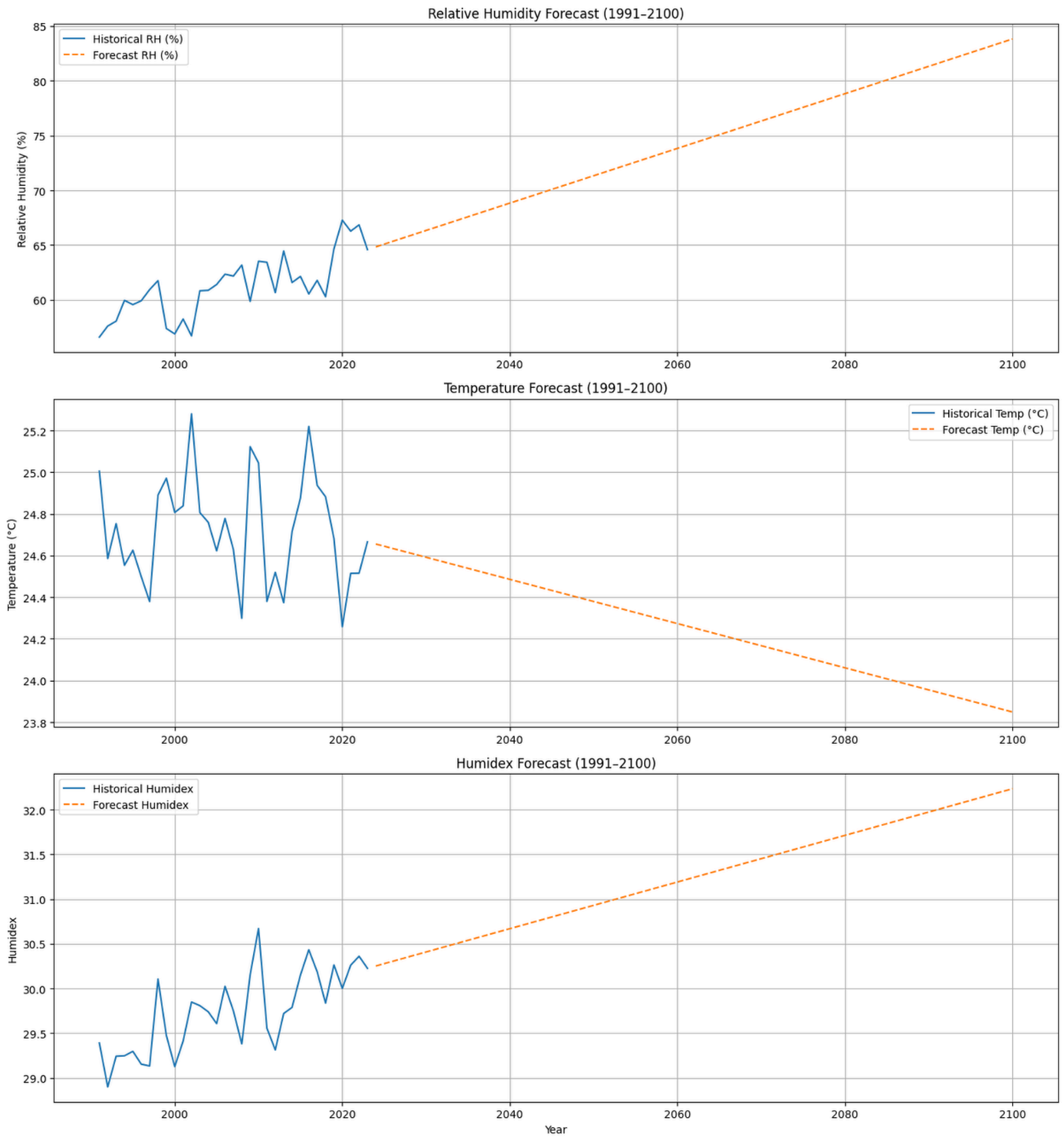
e_s *is the saturation vapor pressure*

T Base = 18 ‘ C

e_{RH} *represents the vapor pressure in pascals and is calculated from RH in percentage and saturated vapor pressure*

Year	India_Temperature	India_Relative_Humidity	Humidex	CDD
1991	25.01	56.61	29.4	4159.53
1992	24.59	57.62	28.9	3980.27
1993	24.75	58.06	29.25	4105.45
1994	24.55	59.96	29.25	4107.42
1995	24.63	59.57	29.3	4125.5
1996	24.5	59.93	29.16	4073.2
1997	24.38	60.94	29.14	4065.86
1998	24.89	61.76	30.11	4421
1999	24.97	57.39	29.48	4190.04
2000	24.81	56.9	29.13	4063.17
2001	24.84	58.26	29.42	4167.62
2002	25.28	56.72	29.85	4326.83
2003	24.81	60.84	29.81	4312.14
2004	24.76	60.88	29.75	4287.08
2005	24.62	61.41	29.61	4238.98
2006	24.78	62.35	30.03	4391.28
2007	24.63	62.18	29.76	4290.78
2008	24.3	63.18	29.39	4156.19
2009	25.12	59.87	30.16	4438.29
2010	25.04	63.53	30.68	4627.44
2011	24.38	63.44	29.56	4220.23
2012	24.52	60.67	29.32	4131.59
2013	24.37	64.47	29.73	4280.34
2014	24.71	61.59	29.79	4305.05
2015	24.88	62.15	30.16	4437.03
2016	25.22	60.55	30.44	4540.32
2017	24.94	61.79	30.19	4450.52
2018	24.88	60.29	29.84	4322.35
2019	24.68	64.65	30.27	4478.68
2020	24.26	67.28	30.01	4383.09
2021	24.51	66.27	30.27	4477.85
2022	24.52	66.85	30.37	4514.32
2023	24.67	64.59	30.23	4464.91


Annual Cooling Degree Days (CDD) in India (1991-2023)
Cooling Degree Days (CDD)
4000
4100
4200
4300
4400
4500
4600
1990
1995
2000
2005
2010
2015
2020
Year

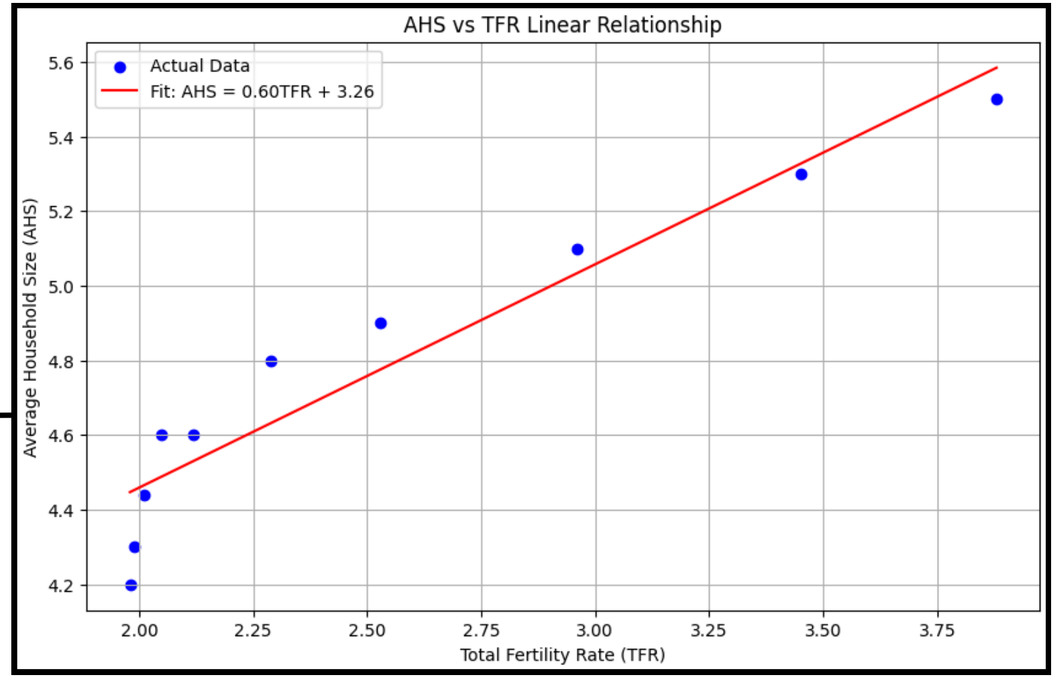


	Year	India_Relative_Humidity	India_Temperature	Humidex
0	2024	64.842136	24.655301	30.252460
1	2025	65.091689	24.644689	30.278569
2	2026	65.341241	24.634078	30.304678
3	2027	65.590793	24.623466	30.330787
4	2028	65.840345	24.612855	30.356896
...	...	...	...	...
72	2096	82.809902	23.891275	32.132306
73	2097	83.059454	23.880663	32.158415
74	2098	83.309006	23.870052	32.184524
75	2099	83.558559	23.859440	32.210633
76	2100	83.808111	23.848829	32.236742

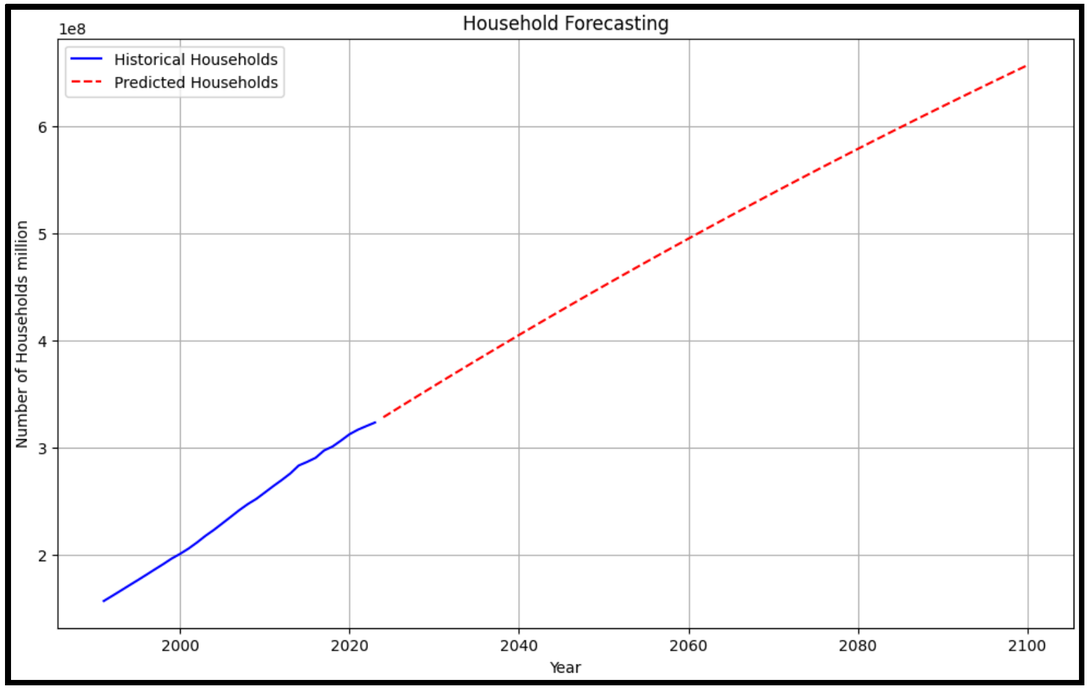
NUMBER OF HOUSEHOLD

$$\frac{POPULATION}{AHS}$$

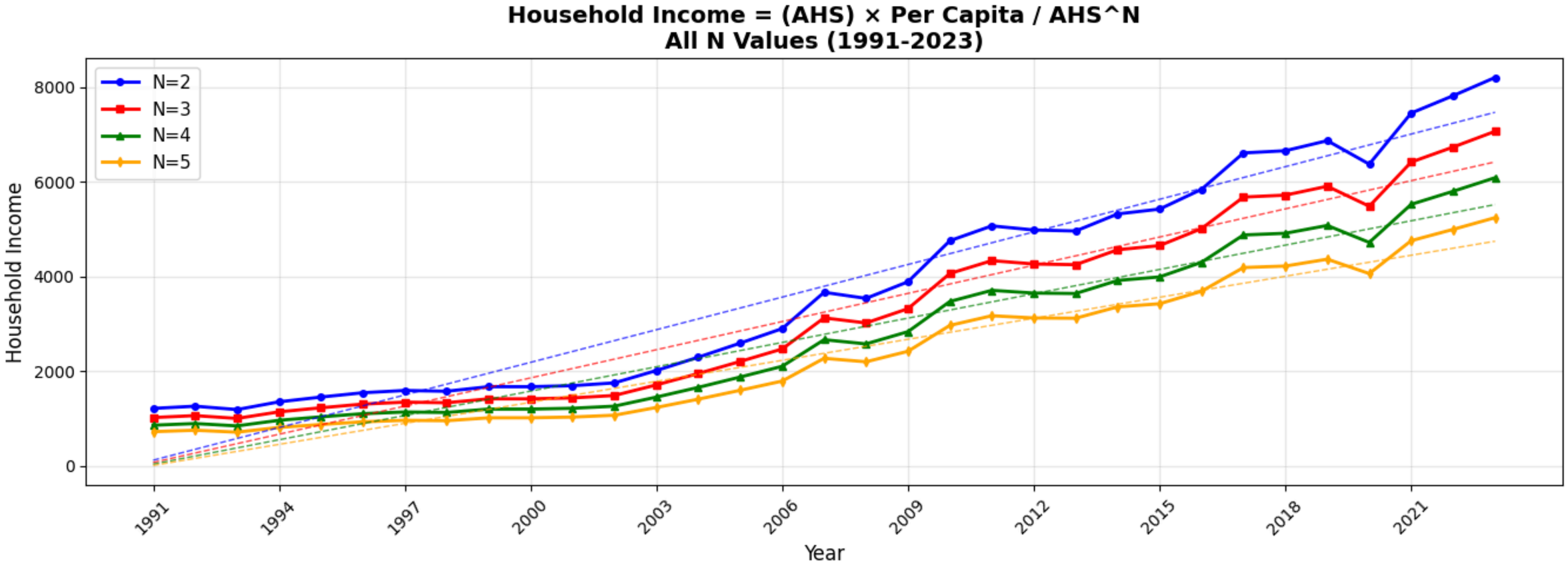
AHS = f(TFR)



AHS = 0.60*TFR + 3.26



HOUSEHOLD FORECAST

HOUSEHOLD INCOME

$$\text{Household Income} = \frac{Average\ House\ Size\ x\ per\ Capita}{(Average\ House\ Size)^{N}}$$